

RUDRA DUDHAT

AI Reliability Engineer. I find the failures AI systems hide, and fix them in production code.

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3rd-year B.Tech in Data Science & AI at IIT Bhilai (CGPA 9.07). 18 bug-fix PRs merged across 12 production repos with 244K combined GitHub stars, and 27 more in review. Seeking AI engineering internships where correctness matters.

OPEN SOURCE

- **18 bug-fix PRs merged** in MLflow, OpenTelemetry, LiteLLM, Kedro, Arize Phoenix, Jaeger, AWS SageMaker SDK, Podman, Strix, Mesa and neural-lam. Each fixes a bug that failed silently, crashed or hung.
- **Telemetry:** MLflow stored agent span data under the wrong keys; OpenTelemetry let NaN values corrupt aggregated metrics and accepted unsorted histogram boundaries; Jaeger panicked on a nil predicate.
- **LLM and cloud tools:** LiteLLM never evicted the last deleted model in multi-instance setups; Arize Phoenix ignored evaluator timeouts when counting retries; SageMaker's stream reader hung without a trailing newline; Podman's VM ready check had no timeout.
- **Bad input:** Kedro rejects project and pipeline names that shadow Python keywords or stdlib modules instead of failing later with confusing import errors; Mesa and neural-lam reject invalid arguments they used to accept silently.

EXPERIENCE

AI Automations Intern, Altagic (performance marketing agency) · Remote

May 2026 – Jul 2026

- Designed and shipped AI and automation systems for e-commerce marketing clients: about 67,500 lines of code across 185 commits in 2 months. Stack: n8n, Shopify Admin API (GraphQL), Slack webhooks, Instantly, Google Sheets, LLM APIs.
- **Email outreach pipeline:** a Google Sheets trigger feeds n8n and Instantly with inbox rotation for high-volume sending, send times tuned to each customer's time zone using Shopify shipping data, and an A/B testing and reply-tracking dashboard.
- **Monitoring and content:** store health monitoring that pulls Shopify data into per-brand Slack channels with three alert tiers (real-time threshold alerts, weekly digest, monthly deep-dive), and an AI pipeline that turns a product list into designed, SEO-optimized HTML pages.

Domain Lead, AI/ML, OpenLake (IIT Bhilai's open source club)

Apr 2026 – Present

- Organizing lead and main speaker for the MLH Hack Day (Gemma 4 Challenge): planned the full-day schedule and two MLH challenges, and lead the 2-hour opening session for about 300 expected participants.
- Mentor students in hands-on AI/ML workshops. Fixed security bugs in the club's Student Database app, including missing login rate limits and an IDOR where profile updates trusted a client-supplied user ID.

RECOMMENDATIONS

"Despite being a first-time intern, he demonstrated a strong ability to learn, take ownership, and deliver quality work independently. He requires minimal supervision, is dependable, and consistently approaches challenges with a positive, solution-oriented mindset."

Shubhangini Thakur, People Ops, who hired and managed me directly · [LinkedIn recommendation](#), June 2026

PROJECTS

build-your-own-llm · Python · in progress

[GitHub](#)

- Writing the AI stack by hand to learn it. My autograd engine matches PyTorch gradients (max difference $7e-15$) on 200 of 200 random graphs. My BPE tokenizer round-trips 1,000 of 1,000 strings and reaches 2.92 bytes per token (5,000-token vocabulary, held-out Shakespeare), 94% of GPT-2's 50K-token tokenizer. 20 tests, 97 to 100% coverage. Transformer, vector database and agent next.

cacheguard · Python, LiteLLM, OpenTelemetry

[GitHub](#) · [write-up](#)

- Semantic caches reuse an old answer when a new question looks similar, and sometimes serve a wrong one with no warning. On 5,000 human-labeled question pairs, about 9% of reused answers are wrong at the commonly recommended 0.85 threshold (95% CI 7.9 to 9.4%, at 70% repeat traffic).
- Built a safety layer that sends borderline matches to an LLM judge: 70% fewer wrong answers (644 to 191, 95% CI 65 to 75%) on 300 pairs while keeping 69% of correct hits. Switching the embedding model to all-mpnet-base-v2 alone cuts wrong answers 17%. 45 tests, 89% coverage.

diffprompt · Python, HDBSCAN

[PyPI v1.2.0](#) · [GitHub](#)

- Teams change a prompt and cannot tell what got worse. Built a CLI that runs both versions on generated test inputs, judges each pair with a cheap model (escalating to a larger one on 4% of calls), and clusters the reasons into named failure modes.
- On 90 labeled prompt pairs, the default CI gate catches 23 of 30 seeded regressions and fails 3 of 45 safe changes (F1 0.82). 95 tests.

brok · Python, MCP, no LLM in the core

[GitHub](#)

- LLMs reviewing system designs are confidently wrong. Built an MCP server that finds the bottleneck and estimates capacity from cited throughput ceilings with deterministic math: identical output across 200 runs, 0.19 ms per review, 359 tests at 98% coverage.
- On 21 designs with answers derived by hand from the cited ceilings, it named the bottleneck 21 of 21 times; GPT-4o-mini did 67% of the time and was off by 100x or more on 18 of 19 capacity estimates. Also a 53-entry cited trade-off knowledge base (93% recall@3 on 30 queries).

OptiQuant · LightGBM, CatBoost, Random Forest, Docker, AWS

[GitHub](#) · [live demo](#)

- End-to-end trading research system: 12+ lookahead-bias-free features, a 3-model ensemble, a backtesting engine and live signal scoring (backtest: 28.16% CAGR, 1.05 Sharpe). Shipped with Docker and GitHub Actions CI/CD on AWS EC2, now on Streamlit Cloud.

SKILLS

ML: Python, PyTorch, scikit-learn, LightGBM, CatBoost, sentence-transformers, HDBSCAN, pytest

LLM, automation and infra: MCP, LangGraph, n8n, LiteLLM, OpenTelemetry, FastAPI, Shopify API, Docker, AWS EC2, GitHub Actions

EDUCATION

Indian Institute of Technology Bhilai · B.Tech in Data Science & Artificial Intelligence

2028 · CGPA 9.07 (through 5th semester)